

VISUAL AUTHENTICATION USING SESSION PASSWORD



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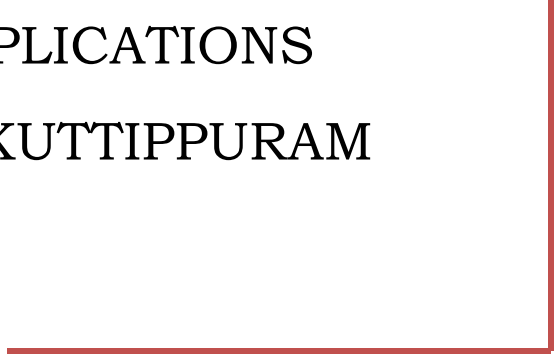


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VISUAL AUTHENTICATION USING SESSION PASSWORD

- Implements a visual authentication protocol using session passwords to enhance online banking security.
- Designed to protect against keylogger attacks, shoulder surfing, eavesdropping, and dictionary attacks.
- Uses two techniques for generating session passwords:
 - Text-based method.
 - Color-based method.



VISUAL AUTHENTICATION USING SESSION PASSWORD

- Provides a role-based banking portal with separate access for Admin, Bank, Employee, and Customer.
- Supports secure account management, loan processing, transaction viewing, customer support, complaints handling, loan requests, notifications, and chat systems.
- Ensures user-friendly yet secure login and transaction processes to improve trust in online banking



OBJECTIVES

- To eliminate vulnerabilities from traditional text-based passwords.
- To provide a role-based access system with distinct functionalities for each user type.
- To streamline banking operations including account management, loan processing and transaction tracking.
- To enable secure and real-time communication between bank staff and customers.
- To maintain detailed logs and transaction histories for transparency and accountability.



EXISTING SYSTEM

- Relies mainly on text-based passwords for authentication.
- Vulnerable to keylogging, phishing, and shoulder surfing attacks.
- Lacks advanced mechanisms to ensure secure online transactions.
- Provides only basic role management without enhanced security.
- Limited support for secure communication and complaint handling.
- Users often face trust issues due to weak authentication methods.



PROPOSED SYSTEM

- Uses visual authentication with session passwords (text & color-based).
- Protects against keylogging, phishing, and shoulder surfing.
- Provides role-based banking portal (Admin, Bank, Employee, Customer).
- Enables secure account, loan, and transaction management.
- Supports complaints, notifications, and chat modules for better service.
- Ensures user-friendly, secure, and trusted online banking.



MOTIVATIONS

- Rising security threats in online banking (keylogging, phishing, shoulder surfing).
- Weaknesses of traditional text-based password systems.
- Need for secure yet user-friendly authentication methods
- Growing demand for trustworthy digital banking platforms.
- Importance of efficient role-based banking operations.
- Enhancing customer satisfaction through secure communication and quick support.

FUNCTIONALITIES

- Admin – view accounts/loans, handle complaints, and send replies.
- Bank - manage employees, accounts, loans, notifications, and customer interactions.
- Employee - Manage customer accounts, update loan requests, deposit funds, and chat with users.
- Customer – Secure login, view account/loan details, transfer money, request loans, send complaints, and chat with bank staff.

FUNCTIONALITIES

- Authentication – Visual session password system to secure logins and transactions.
- Communication – Built-in chat and complaint modules for quick support.
- Notifications – Banks can publish updates and alerts for customers.

MODULE DESCRIPTION

Admin:

- Login
- Manage Bank
- View account and customers
- View loan details
- View accounts
- View loan requests and its status
- View Complaint
- Send Reply



MODULE DESCRIPTION

Bank:

- Login
- Manage Employee
- View account and customers
- Manage loan details
- Manage Notification
- View accounts
- View loan requests and its status
- Chat with Employee



MODULE DESCRIPTION

Employee:

- Login
- View profile
- Manage account and customers
- View loan requests
- Update status
- Deposit to account
- Chat with Bank
- Chat with Customer



MODULE DESCRIPTION

Customer:

- Secure login
- View account details
- Transfer money to another account
- View all transactions
- View loan details
- Request for a loan
- Check loan request status
- Chat with Employee
- Send Complaint
- View Reply



DEVELOPING ENVIRONMENT

- Operating System – Windows
- Front End – HTML, CSS, JavaScript
- Back End – Python
- Framework – Flask
- Database – MySQL
- IDE / Code Editor – Visual Studio Code (VS Code)
- Web Server – wamp



SPRINT BACKLOG

Backlog tem	Status And Completion Date	Original Estimatio n in Hours	Day 1 hrs	Day 2 hrs	Day 3 hrs	Day 4 hrs	Day 5 hrs	Day 6 hrs	Day 7 hrs	Day 8 hrs	Day 9 hrs	Day 10 hrs
SPRINT1												
Table designing	14/8/2025	2	1	1	0	0	0	0	0	0	0	0
Form designing	16/8/2025	3	1	1	1	0	0	0	0	0	0	0
Coding	18/8/2025	3	1	1	1	0	0	0	0	0	0	0
SPRINT 2												
Database schema collection	21\8/2025	7	1	1	1	1	1	1	1	0	0	0
admin dashboard coding	24/8/2025	7	1	1	1	1	1	1	1	0	0	0



SPRINT BACKLOG

Backlog tem	Status And Completion Date	Original Estimatio n in Hours	Day 1 hrs	Day 2 hrs	Day 3 hrs	Day 4 hrs	Day 5 hrs	Day 6 hrs	Day 7 hrs	Day 8 hrs	Day 9 hrs	Day 10 hrs
SPRINT3												
account managemen t	28/8/2025	7	1	1	1	1	1	1	1	0	0	0
loan handling module	1/9/2025	7	1	1	1	1	1	1	1	0	0	0
SPRINT 4												
testing	27/9/2025	7	1	1	1	1	1	1	1	0	0	0
reports and certificates	10/10/2025	7	1	1	1	1	1	1	1	0	0	0
TOTAL		50	9	9	8	6	6	6	6			



PRODUCT BACKLOG

ID	NAME	PRIORITY <high/medium/low>	ESTIMATE (Hours)	STATUS <Planned/In progress/Completed>
1	ADMIN LOGIN AND MANAGMENT	High	3	COMPLETED
2	VIEW ACCOUNT AND CUSTOMERS	MEDIUM	2	COMPLETED
3	VIEW LOAN DETAILS AND REQUEST	MEDIUM	2	COMPLETED
4	VIEW LOAN DETAILS AND REQUEST	MEDIUM	2	COMPLETED
5	HANDLE COMPLAINTS	MEDIUM	2	COMPLETED

PRODUCT BACKLOG

ID	NAME	PRIORITY <high/medium/low>	ESTIMATE (Hours)	STATUS <Planned/In progress/Completed>
6	BANK LOGIN AND MANAGMENT	High	3	COMPLETED
7	MANAGE EMPLOYEES	MEDIUM	2	COMPLETED
8	MANAGE NOTIFICATIONS	MEDIUM	1	COMPLETED
9	EMPLOYEE LOGIN AND MANAGMENT	HIGH	2	COMPLETED
10	MANAGE CUSTOMER ACCOUNT	HIGH	2	COMPLETED

PRODUCT BACKLOG

ID	NAME	PRIORITY <high/medium/low>	ESTIMATE (Hours)	STATUS <Planned/In progress/Completed>
11	DEPOSIT TO CUSTOMER ACCOUNT	MEDIUM	2	COMPLETED
12	CHAT WITH BANK AND CUSTOMER	LOW	1	COMPLETED
13	CUSTOMER LOGIN (SESSION PASSWORD)	HIGH	3	COMPLETED
14	VIEW ACCOUNT DETAILS AND TRANSACTION	HIGH	2	COMPLETED
15	TRANSFER MONEY TO OTHER ACCOUNTS	HIGH	3	COMPLETED

USER STORY

User Story ID	As a type of User	I want to <Perform some task>	So that i can <Achieve Some Goal>
1	ADMIN LOGIN AND MANAGEMENT	Login	login successful and access system securely
2	ADMIN	View and Manage	maintain bank with customer data
3	ADMIN	view loan	approve or reject loan
4	BANK	Login	login successful with correct username and password
5	BANK	managing	employees, loan, notification
6	BANK	chat with employees	quick staff support
7	employee	login	access system

PROJECT PLAN

User StoryID	Task Name	Start Date	End Date	Days	Status
1	Sprint 1	14/08/2025	15/08/2025	10	COMPLETED
4		16/08/2025	17/08/2025		COMPLETED
5		18/08/2025	19/08/2025		COMPLETED
10		20/08/2025	21/08/2025		COMPLETED
11		22/08/2025	23/08/2025		COMPLETED
2	Sprint 2	3/09/2025	4/09/2025	16	COMPLETED
7		5/09/2025	6/09/2025		COMPLETED
12		7/09/2025	8/09/2025		COMPLETED

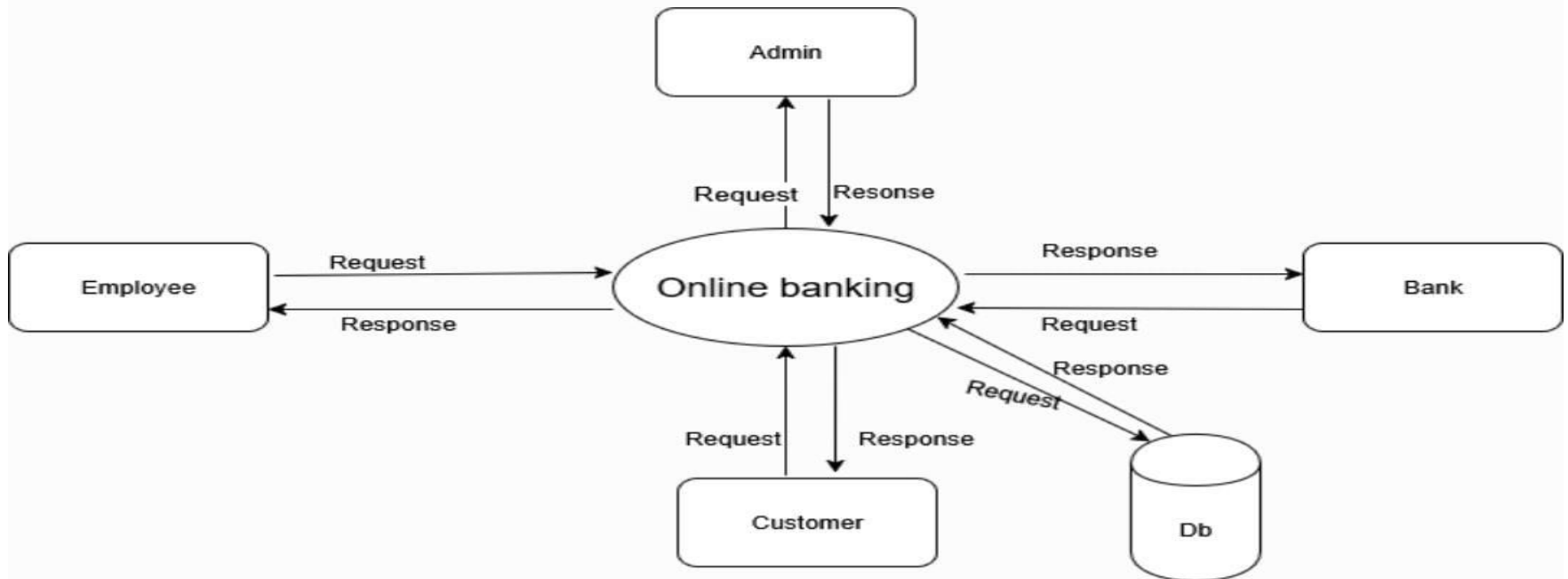


PROJECT PLAN

User StoryID	Task Name	Start Date	End Date	Days	Status
3	SPRINT 3	9/9/2025	10/9/2025	14	COMPLETED
6		11/9/2025	14/9/2025		COMPLETED
13		15/9/2025	19/9/2025		COMPLETED
14	SPRINT 4	27/9/2025	28/9/2025	10	COMPLETED
8		29/9/2025	30/9/2025		COMPLETED
9		5/10/2025	10/10/2025		COMPLETED

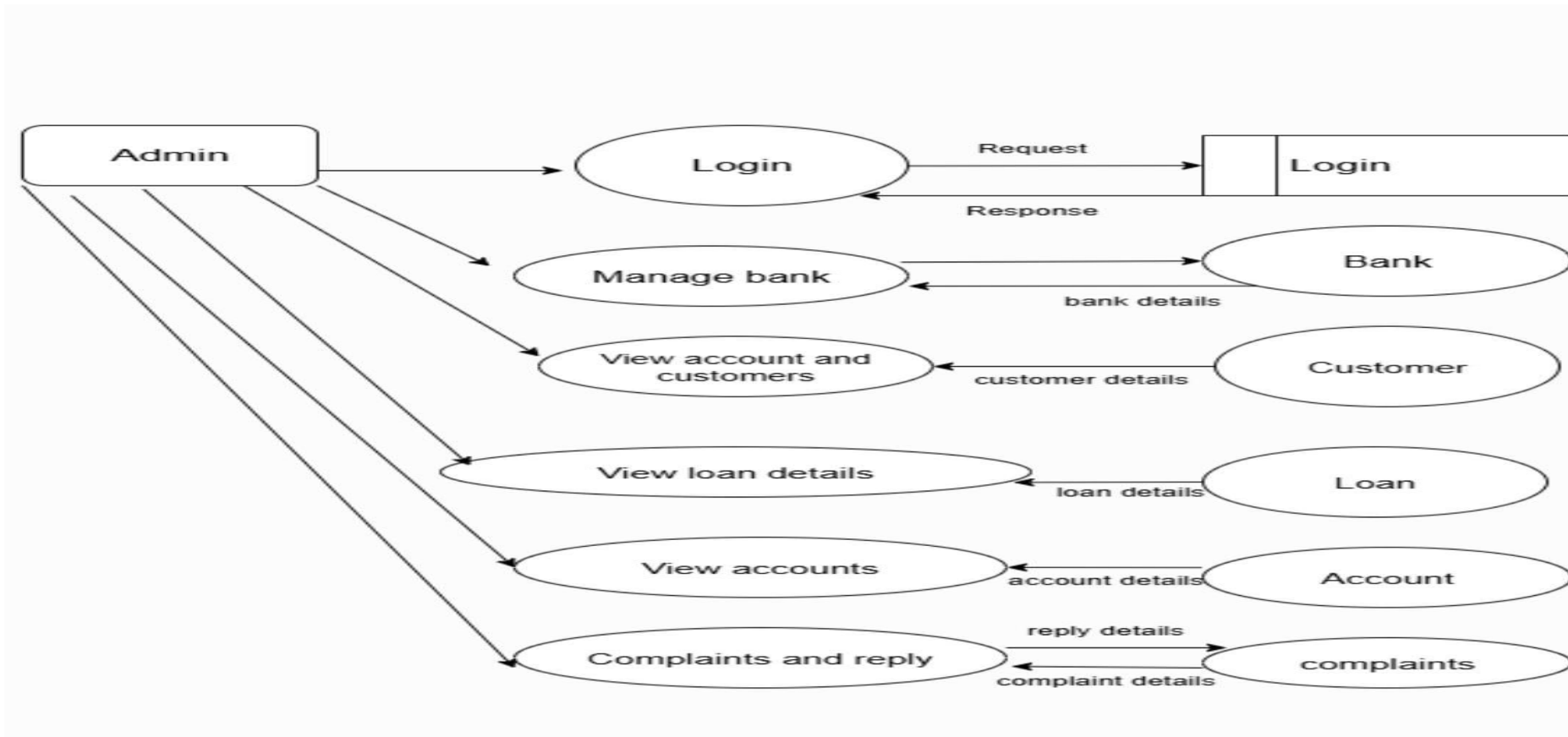
DATA FLOW DIAGRAM

- LEVEL 0

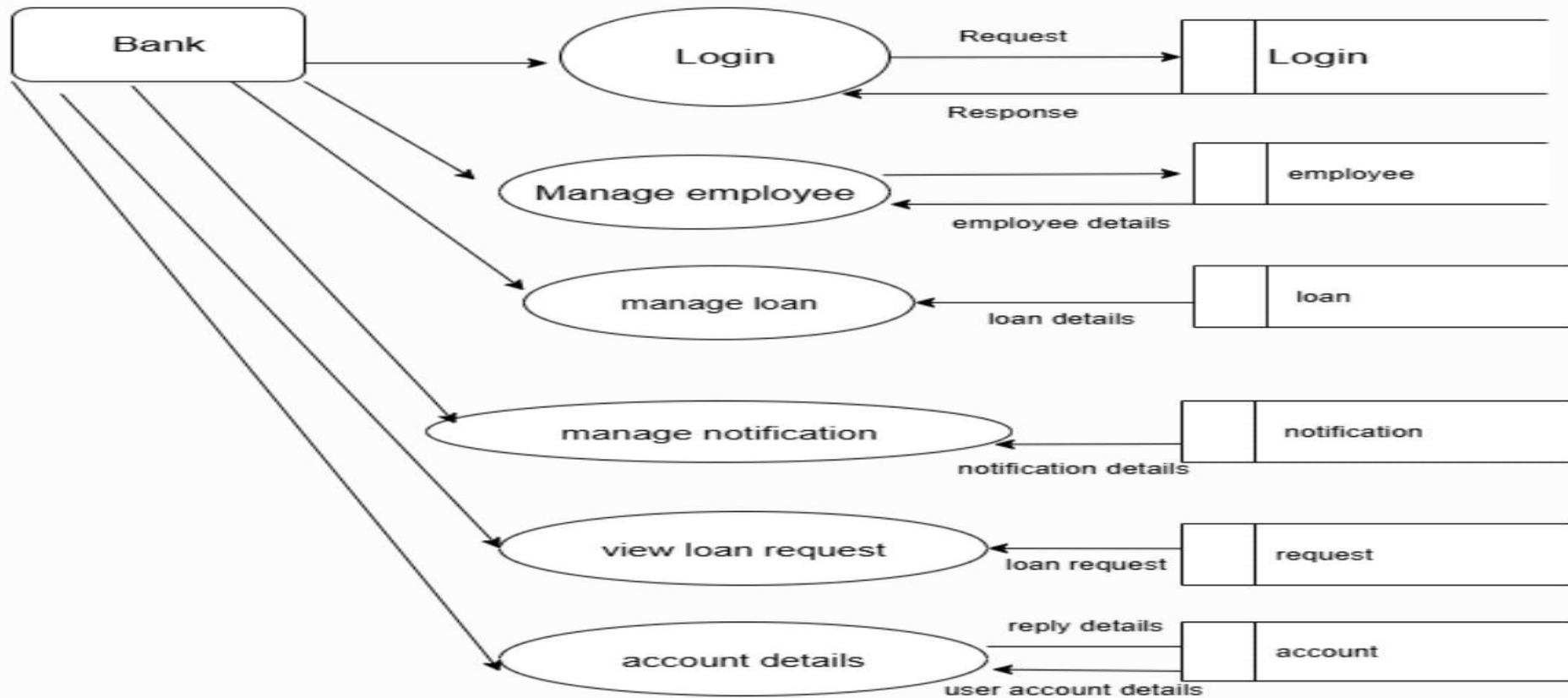


DATA FLOW DIAGRAM

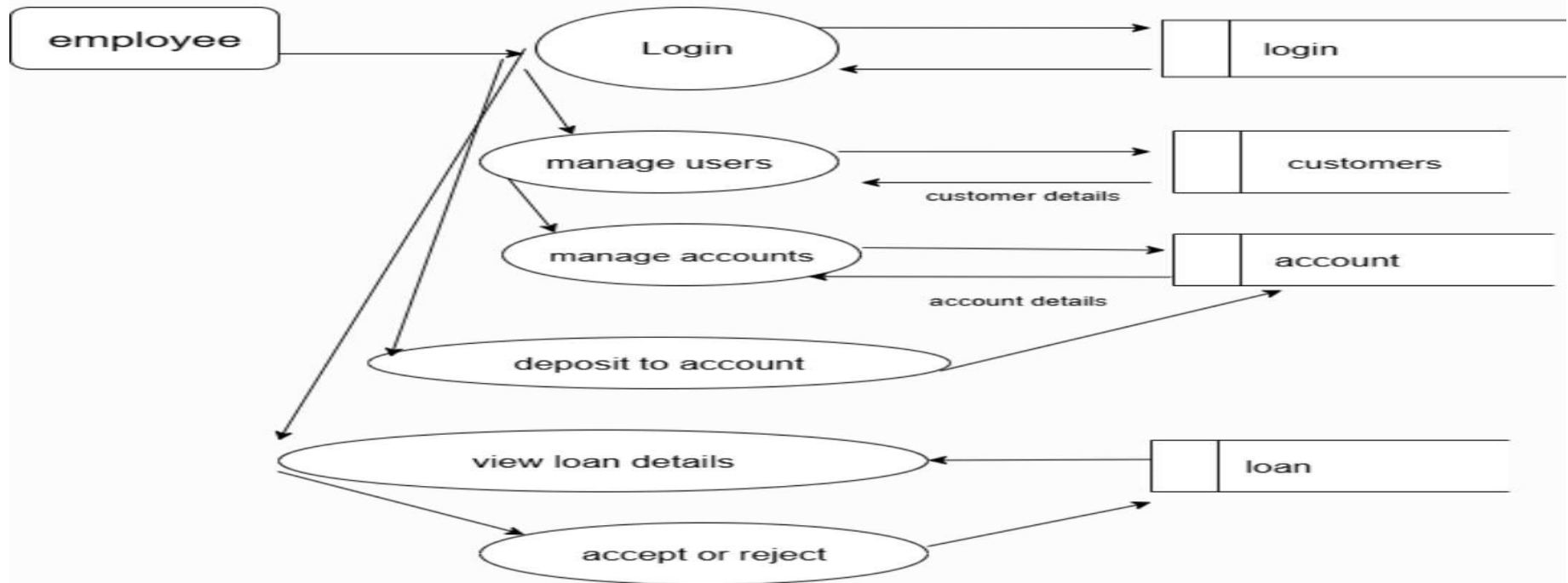
- Level 1



• LEVEL 2

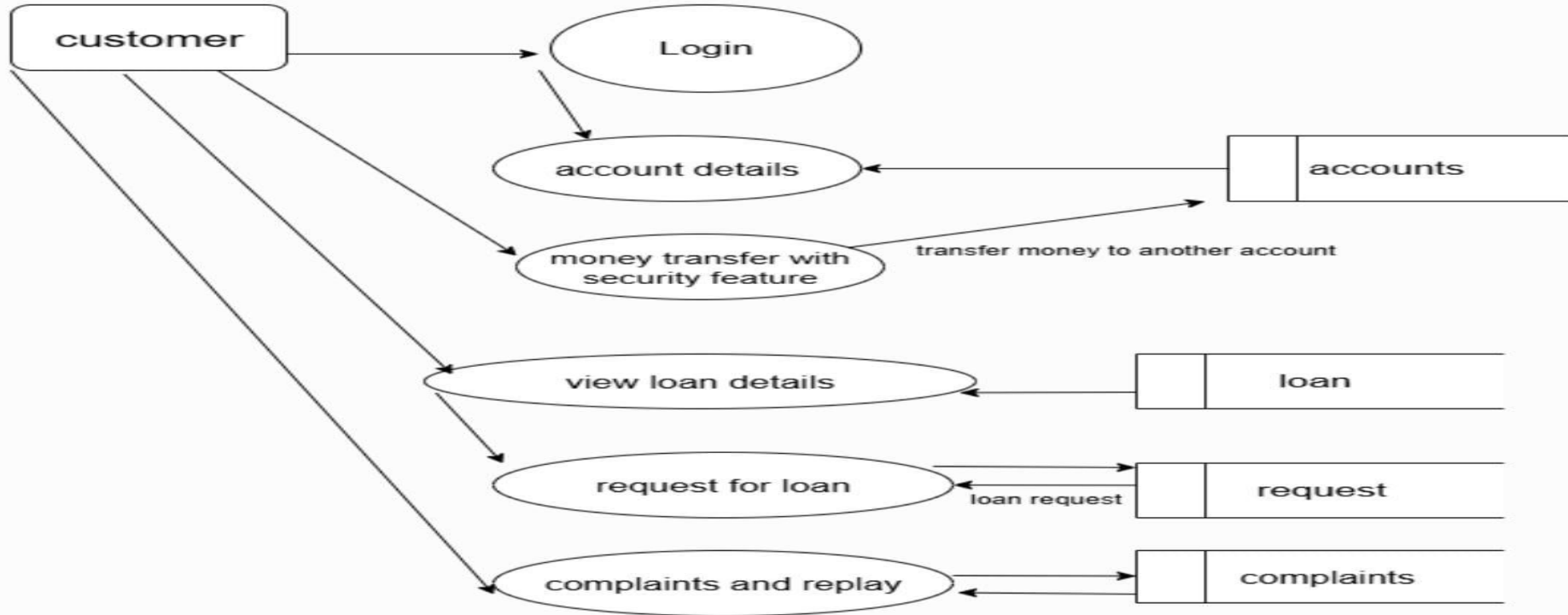


• LEVEL 3

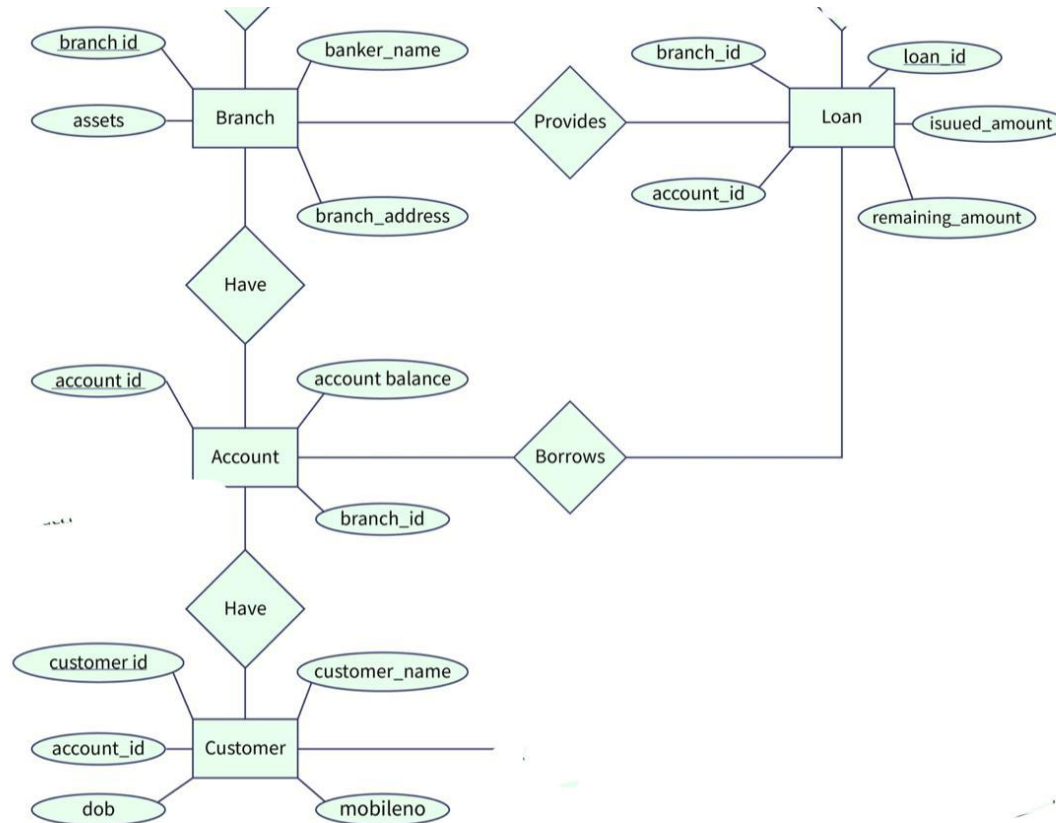


DATA FLOW DIAGRAM

- Level 4



ER DIAGRAM



Code:

- `from flask import *`
- `from database import *`
- `import random`

- `import smtplib`
- `from email.mime.text import MIMEText`
- `from flask_mail import Mail`

- `customer=Blueprint('customer',__name__)`

- `@customer.route('/customerhome')`
- `def customerhome():`
- `data={}`
- `data['cane']=session['cname']`
- `return render_template("customerhome.html",data=data)`

- `@customer.route('/customerviewaccountdetails')`
- `def customerviewaccountdetails():`
- `cid=session['cid']`
- `data={}`
- `q="select * from account inner join bank using(bank_id) where customer_id='%s'"%(cid)`
- `res=select(q)`
- `data['account']=res`

- `return render_template("customerviewaccountdetails.html",data=data)`

```

• @customer.route('/customersendcomplaint',methods=['get','post'])
• def customersendcomplaint():
•     cid=session['cid']
•     data={}
•     q="select * from complaint where customer_id='%s'"%(cid)
•     res=select(q)
•     data['complaint']=res
•     if 'send' in request.form:
•         complaint=request.form['complaint']
•         q="insert into complaint values(null,'%s','%s','pending',curdate())"%(cid,complaint)
•         insert(q)
•         return redirect(url_for('customer.customersendcomplaint'))

•     return render_template("customersendcomplaint.html",data=data)

• @customer.route('/customerviewloandetails',methods=['get','post'])
• def customerviewloandetails():
•     cid=session['cid']
•     data={}

•     q="select * from loans inner join bank using(bank_id) where bank_id in (select bank_id from account where customer_id='%s' and
status='Accepted')"%(cid)
•     res=select(q)
•     if res:
•         data['loans']=res
•     else:
•         flash("YOU HAVE NO ACCOUNTS")
•         return redirect(url_for('customer.customer_view_banks'))

•     if 'action' in request.args:
•         action=request.args['action']
•         lnid=request.args['lnid']
•         q="insert into loanrequest values(null,'%s','%s',now(),'pending')"%(cid,lnid)
•         insert(q)

```

```

• original=""
•
• c1=a.index('violet')
• c2=a.index('indigo')
• c3=a.index('blue')
• c4=a.index('green')
• c5=a.index('yellow')
• c6=a.index('orange')

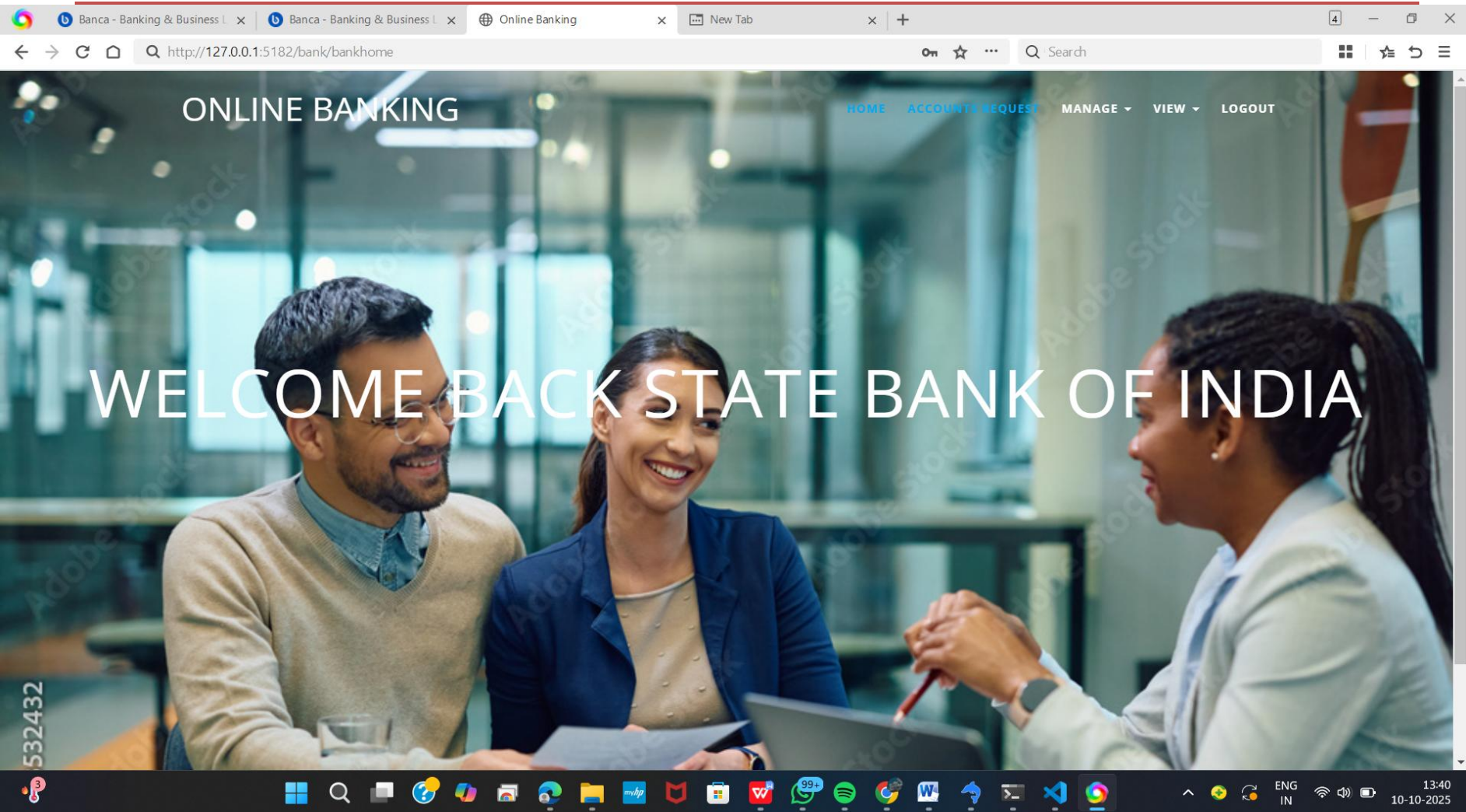
•
• original=str(v[c1])+str(v[c2])+str(v[c3])+str(v[c4])+str(v[c5])+str(v[c6])
• print("oooo",original)

•
• q="select * from account where image='%s' and account_number='%s'"%(original,ac)
• print(q)
• res=select(q)
•
• print("rrr",res)
• if res:
•
•     insert(session['trans1'])
•     update(session['trans2'])
•     update(session['trans3'])
•     flash("Money sended successfully")
•     return redirect(url_for('customer.customerhome'))
•

```

-
- `@customer.route('/customerviewalltransations')`
 - `def customerviewalltransations():`
 - `data={}`
 - `cid=session['cid']`
 - `q="select * from bank where bank_id in (select bank_id from account where`
`customer_id='%s' and status='Accepted')"% (cid)`
 - `res=select(q)`
 - `if res:`
 - `data['loans']=res`
 - `else:`
 - `flash("YOU HAVE NO ACCOUNTS")`
 - `return redirect(url_for('customer.customer_view_banks'))`
 - `q="select * from transactions where from_ac in (select account_number from account`
`where customer_id='%s') or to_ac in (select account_number from account where`
`customer_id='%s')"% (cid,cid)`
 - `res=select(q)`
 - `data['transactions']=res`
 - `return render_template("customerviewalltransations.html",data=data)`

INTERFACES



Browser tabs: Banca - Banking & Business L x Banca - Banking & Business L x New Tab

Address bar: http://127.0.0.1:5180/login

Navigation: HOME LOGIN NEW USER

Smart finance solutions for your business

Inventore veritatis et architecto beatae vitae dicta explicabo nemo enim ipsam voluptatem quia voluptas aspernatur magni .

Login Here

User Name

Password

Sign IN

New User Registration [Click Here](#)

Windows taskbar: Search, 99+ notifications, 10:56 23-09-2025



FUNCTIONALITIES

The screenshot displays a web browser window with the URL `http://127.0.0.1:5182/customer/customertransfermoneytoanotheraccount`. The page title is "ONLINE BANKING". The navigation menu includes "HOME", "VIEW", "CHECK REQUEST STATUS", "TRANSFER MONEY", "COMPLAINT", and "LOGOUT". The "TRANSFER MONEY" section contains the following form fields:

TRANSFER MONEY	
FROM ACCOUNT	758469896156
TO ACCOUNT	503601659027
AMOUNT	100
TRANSFER	

The background of the page features a dark, abstract design with a faint "VISA" logo and a stylized "S" logo. The Windows taskbar at the bottom shows the time as 11:15 on 08-10-2025.



Online Banking

HOME VIEW CHECK REQUEST STATUS TRANSFER MONEY COMPLAINT LOGOUT

Key

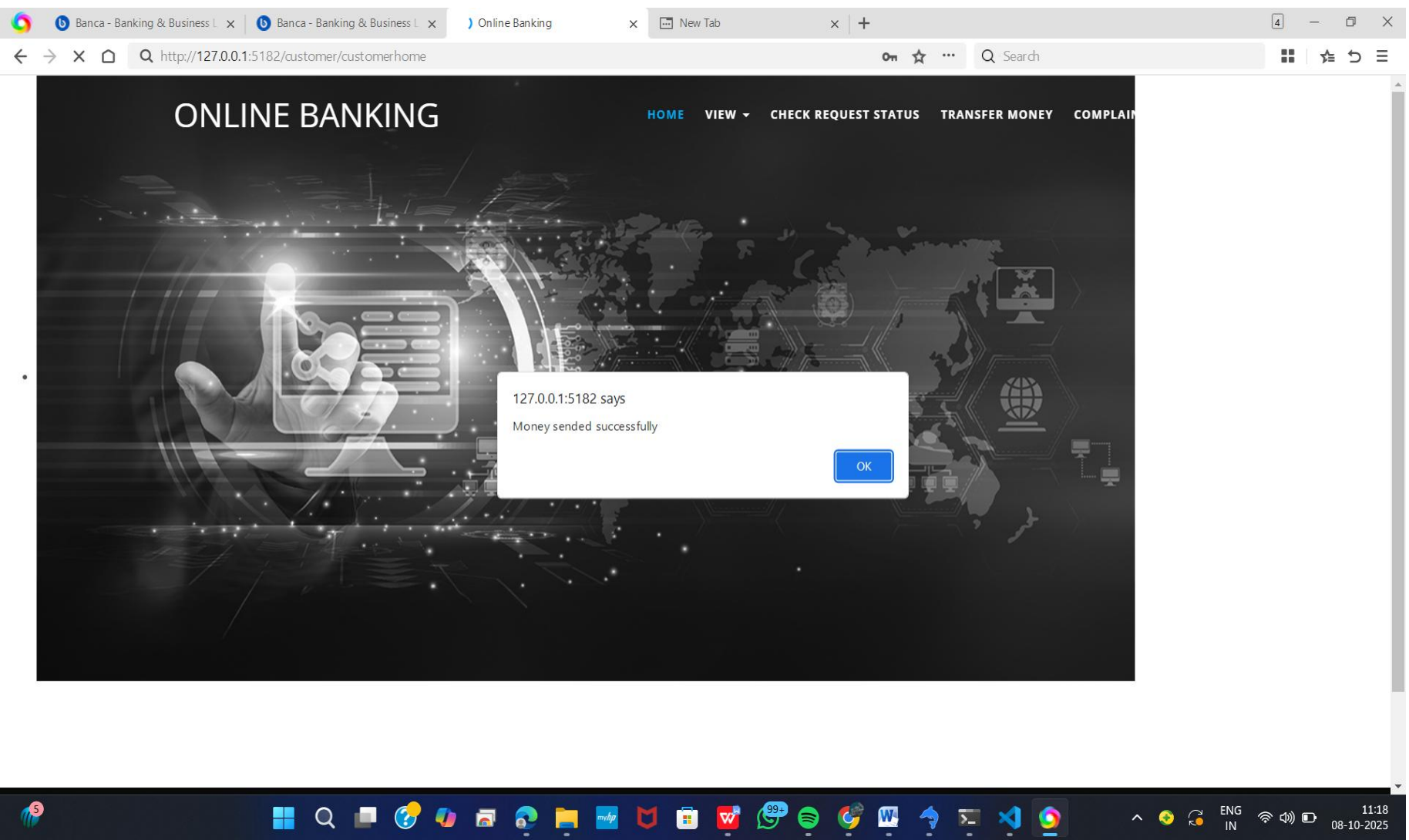
21	7	13	14	30	2
5	35	3	28	36	26
27	10	8	32	34	25
29	24	12	15	0	33
16	6	1	23	11	22
19	31	18	4	17	20

34,26,29

VERIFY

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ONLINE BANKING

Transaction Receipts

Search From Account Search To Account dd-mm-yyyy

Payment Successful
FED0877

FROM ACCOUNT 758469896156 TO ACCOUNT 503601659027

Amount Transferred
INR 2000.00

2025-10-06 17:53:35 COMPLETED

Payment Successful
FED8570

FROM ACCOUNT 758469896156 TO ACCOUNT 503601659027

Amount Transferred
INR 300.00

2025-10-06 17:59:22 COMPLETED

Payment Successful
FED2067

FROM ACCOUNT 758469896156 TO ACCOUNT 503601659027

Amount Transferred
INR 100.00

2025-10-08 11:18:00 COMPLETED



THANK YOU